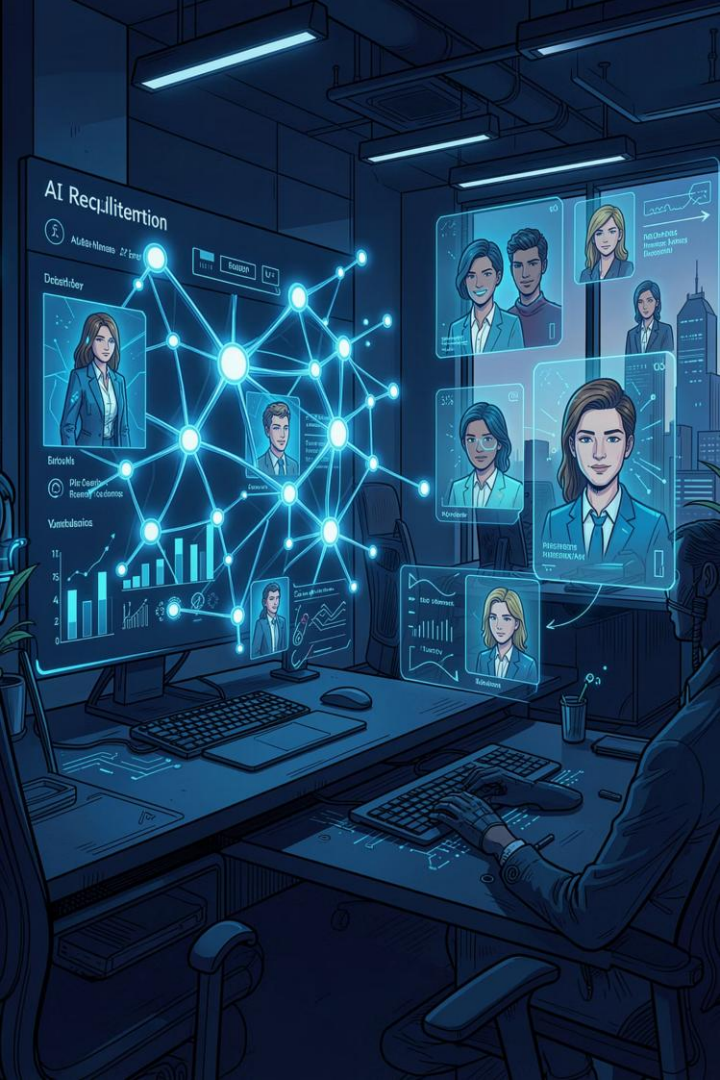




# HR Resume & LinkedIn Shortlisting Agent

*AI Enablement Internship Project*

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# The Problem with Manual Screening

## 🕒 Time-Consuming

HR teams manually screen **hundreds of resumes** per role, consuming days of productive time.

## 📉 Inconsistent

Evaluation quality degrades with volume — no two reviewers apply the same standard.

## 🧠 Bias-Prone

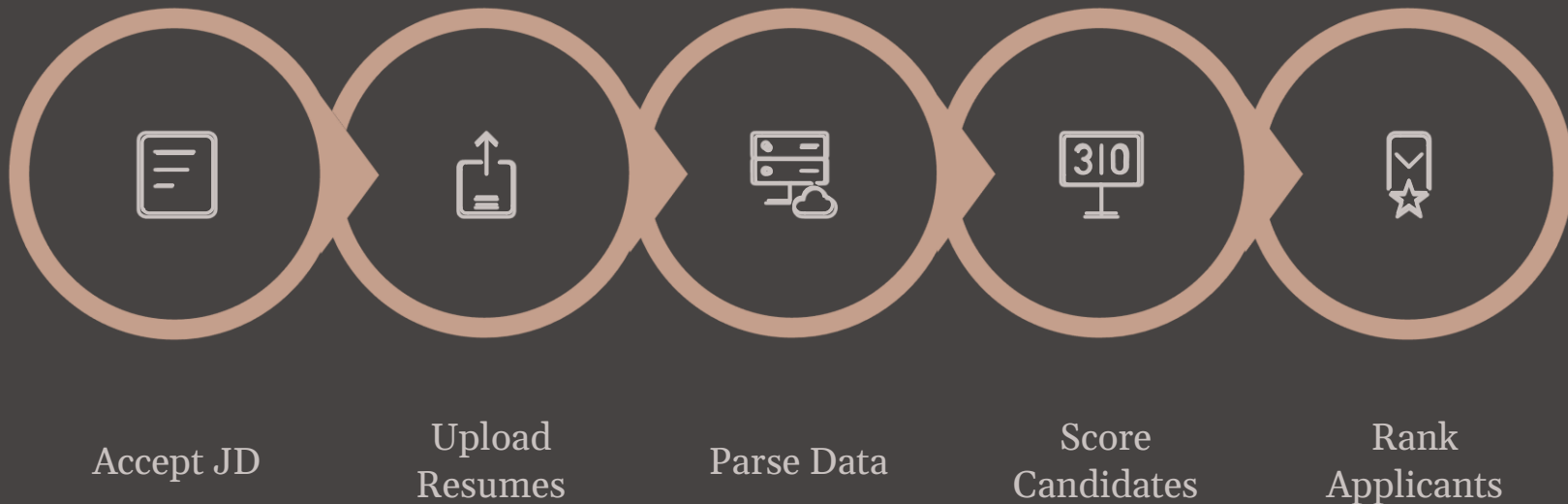
Human fatigue introduces **unconscious bias**, risking unfair and legally questionable decisions.

## 🔔 Scalability Gap

Growing hiring needs demand an **intelligent, automated** shortlisting solution.

# Project Overview

An end-to-end AI-powered pipeline that ingests a job description and a batch of resumes, then automatically scores, ranks, and reports on every candidate — with HR staying in control.



From raw files to a downloadable ranked report, the system handles every step while preserving a human-in-the-loop override at the final stage.

# Features Implemented



## JD & Resume Parser

Extracts key requirements from job descriptions and parses PDF/DOCX candidate files automatically.



## Ranking System

Automatically orders all applicants by composite score for instant shortlist visibility.



## HTML Report Generator

Produces a styled, downloadable report with candidate scorecards via Jinja2 templates.



## Candidate Scoring Engine

Applies a weighted rubric across five dimensions to generate a normalized candidate score.



## Human-in-the-Loop Override

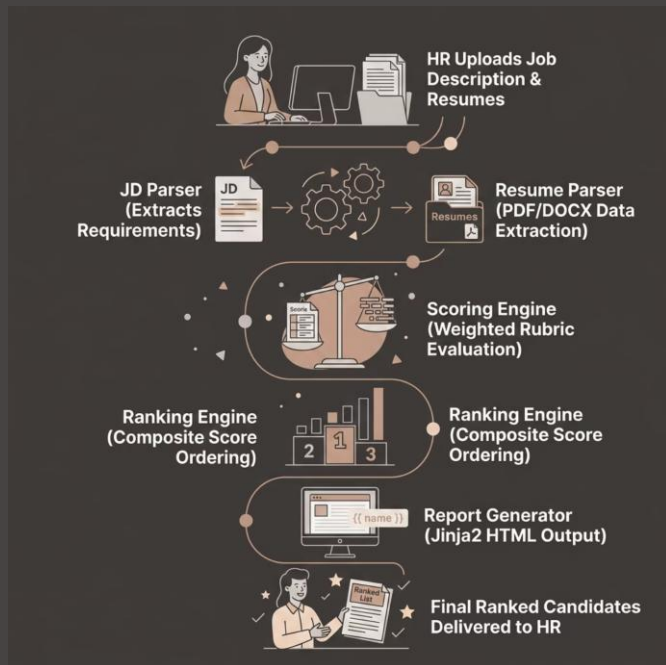
Recruiters can manually adjust rankings before any hiring decision is finalized.



## Streamlit UI

A clean, browser-based interface requiring zero setup — accessible to any HR team member.

# System Architecture



## How It All Connects

The pipeline is fully sequential and stateless — each module hands off structured JSON data to the next, ensuring transparency and easy debugging.

### Rule-Based Logic

No LLM dependency — deterministic and auditable at every step.

### Local Execution

All processing runs on-device. No data leaves the environment.

# Candidate Scoring Rubric

| Scoring Dimension          | Weight | Priority                                                                                    |
|----------------------------|--------|---------------------------------------------------------------------------------------------|
| Skills Match               | 30%    |  High      |
| Experience Relevance       | 25%    |  High      |
| Projects & Portfolio       | 20%    | <br>Medium |
| Education & Certifications | 15%    | Medium                                                                                      |
| Communication Quality      | 10%    | Low                                                                                         |

## Why Weighted Scoring?

Each dimension is scored independently and multiplied by its weight, producing a **0–100 composite score**. This ensures roles with different priorities can be served by simply adjusting weights — no code changes needed.

 The rubric is fully configurable per job description, giving HR teams complete control over evaluation criteria.

# Tech Stack

## Core Language

Python · pdfplumber · python-docx · JSON



## Interface

Streamlit — zero-config browser UI for HR teams



## Reporting

Jinja2 templates · HTML report generation



## Architecture

Rule-based AI · No API key required · Fully local & lightweight



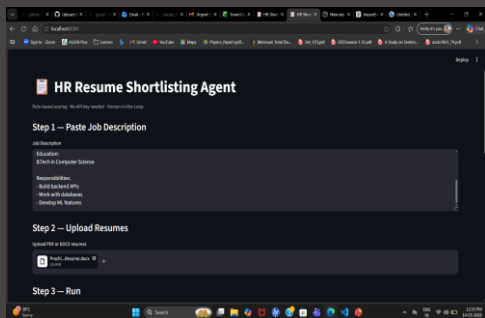
# Security & Privacy Mitigations



## Built-In Safeguards

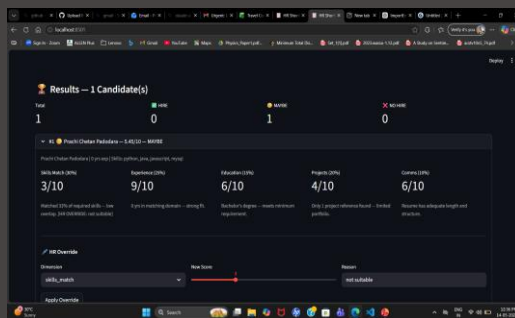
- **File type restrictions** — only PDF and DOCX accepted; malicious uploads blocked at intake.
- **Local-only processing** — no candidate data transmitted to external servers or APIs.
- **No hardcoded API keys** — eliminates credential leakage risk entirely.
- **Temporary file handling + prompt injection protection** guard runtime integrity.
- **Human review gate** before any final hiring decision is actioned.

# Demo & Results



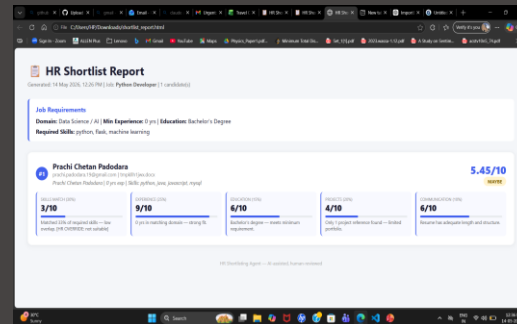
## Streamlit Upload UI

Drag-and-drop resume uploads alongside JD input — ready in seconds with no setup.



## Ranked Candidate Table

All applicants sorted by weighted composite score — top talent surfaces instantly.



## Downloadable HTML Report

Auto-generated scorecards with per-dimension breakdowns — shareable with one click.

# Conclusion & Future Scope

## What We Achieved

### 1 Faster Hiring Workflow

Screening time reduced from days to minutes for large applicant pools.

### 2 Transparent Evaluation

Every score is explainable, auditable, and bias-resistant by design.

### 3 Recruiter Empowerment

HR retains full override authority — AI assists, humans decide.

## What Comes Next



### LinkedIn Integration

Auto-fetch public profiles to supplement resume data.



### LLM Semantic Matching

OpenAI / Gemini support for deeper contextual scoring.



### ATS & Cloud Deployment

Plug directly into enterprise ATS platforms via API.

Thank You